Section 1.6 — Composition of Functions

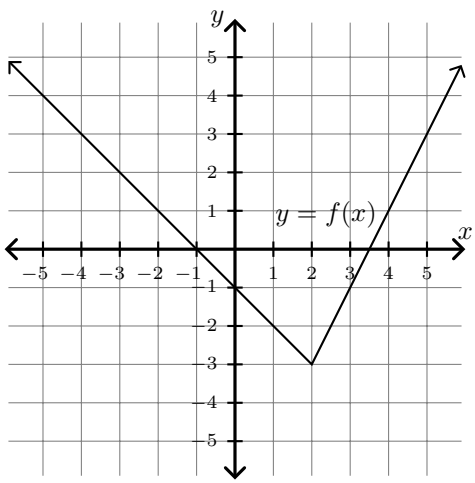
Math 103: College Algebra

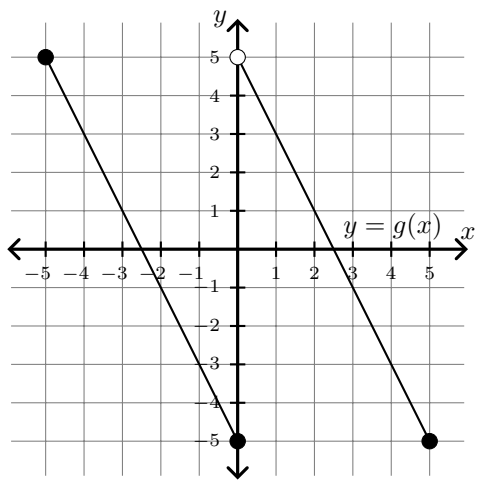
1 Composing Functions

Let $f = \{(1,5),(2,3),(3,1),(4,4),(5,2)\}$ and $g(x) = \{(1,1),(2,4),(3,2),(4,5),(5,3)\}$. Find each of the following.

1. $(f \circ g)(4)$
2. $(g \circ f)(2)$
3. $(f \circ g)(1)$
4. $(f \circ g \circ f)(3)$
5. $(f \circ f)(5)$
6. Write $f \circ g$ as a set of ordered pairs

Let $f(x)$ and $g(x)$ be the functions graphed below.





Find each of the following.

7. $(f \circ g)(-1)$
8. $(g \circ f)(-1)$
9. $(g \circ f)(3)$
10. $(f \circ f)(-2)$

Let $f(x) = 2x - 4$, $g(x) = -\frac{2}{3}x + 1$, and $h(x) = |x + 2|$. Find each of the following.

11. $(f \circ g)(12)$
12. $(h \circ g)(0)$
13. $(h \circ f)(-1)$
14. $(f \circ f)(1)$
15. $(g \circ h)(1)$
16. Write an expression for $(f \circ g)(x)$. Simplify your answer as much as possible.
17. Write an expression for $(g \circ h \circ f)(x)$. Do not simplify your answer.

2 Inverses

Let $f(x) = \frac{1}{2}x - 4$, $g(x) = -2x + 4$, $h(x) = -\frac{1}{2}x + 2$, $k(x) = 2x + 2$. Give expressions for each of the following:

18. $(f \circ g)(x)$
19. $(f \circ h)(x)$
20. $(f \circ k)(x)$
21. $(g \circ h)(x)$
22. $(g \circ k)(x)$
23. $(h \circ k)(x)$
24. Which pair of functions are inverses of one another?

3 Undoing composition

Give functions so that the given function is the given composition. Each function should only do one thing.

25. $(f \circ g)(x) = 2x - 4$
26. $(f \circ g)(x) = |x + 1|$
27. $(f \circ g)(x) = x^2 + 4$
28. $(f \circ g)(x) = (x - 3)^4$
29. $(f \circ g)(x) = 3|x|$
30. $(f \circ g)(x) = \sqrt{x - 5}$

4 Domain and Range

For each pair of functions f and g, find the domain and range of $f \circ g$.

31. $f = \{(-3,5),(0,4),(2,9),(5,4)\}$
    $g = \{(3,5),(5,1),(7,-1),(9,5)\}$

32. $f = \{(-9,-7),(-7,7),(-1,1),(7,1)\}$
    $g = \{(-7,-1),(1,5),(3,-1),(9,7)\}$

33. $f = \{(-9,-8),(-4,3),(5,-3),(6,6)\}$
    $g = \{(-8,7),(-3,-5),(0,-5),(3,7)\}$

34. $f = \{(-7,-7),(-6,0),(-5,-2),(5,8)\}$
    $g = \{(-7,5),(-2,-7),(0,3),(8,1)\}$

For each pair of functions f and g, find the domain of $f \circ g$.

35. $f(x) = \frac{1}{x},\ x > 0$
    $g(x) = x + 4,\ x > 0$

36. $f(x) = \sqrt{x}$ (given $\operatorname{Dom} f = [0, \infty)$)
    $g(x) = -4x + 10,\ x \geq 0$